ALBERTO BOTTARI

Software Engineer | C++ | Systems & Performance
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PROFILE

Software engineer focused on C++, systems, and performance, with a master's degree in Computer Engineering (110/110 cum laude). Hands-on experience with MLIR-based compilation for compute-near-HBM architectures, optimization algorithms, distributed infrastructure, and production software. AI enthusiast, actively exploring AI-assisted software engineering through coding assistants, MCP, and plugins.

WORK EXPERIENCE

imec – TACOS Group Feb 2026 – Aug 2026
Research Internship and Master's Thesis Leuven, Belgium

- Designed and implemented a modular FIMDRAM backend in the MLIR-based Cinnamon compiler for two compute-near-HBM architectures.
- Added a FIMDRAM dialect and intermediate representation, declarative target models, and interchangeable policies for instruction selection, addressing, register allocation, placement, and binary encoding.
- Built staged verification and reproducible compiler artifacts to compare Samsung AAM and a modified host-driven FIMDRAM model across representative workloads.

Freelance Software Development 2023 – Present
Full-stack Developer Italy

- Designed and delivered MedjugorjeService, a multilingual platform that manages radio-loan orders, generates branded PDF documents, and provides an administrative workflow and technical SEO.
- Built SoleAzzurroCucce, an e-commerce platform with email and social authentication, Nexi payments, and administration for orders, products, and users.
- Developed Capricciosa Pizzerie, a production website with managed menus, events, and a dedicated administration panel.

EDUCATION

University of Padua Oct 2024 – Oct 2026
M.Sc. in Computer Engineering Padua, Italy

- Web Information and Data Engineering curriculum. Graduated 110/110 cum laude on 13 October 2026.
- Master's thesis: Modular Compiler Design for Emerging Compute-near-HBM Architectures.

University of Padua Oct 2021 – Jul 2024
B.Sc. in Computer Engineering Padua, Italy

- Graduated 110/110 cum laude. Thesis: Design and Testing of a Flexible Pressure Sensor for Smart Walkers.

RESEARCH AND SYSTEMS PROJECTS

WebSDR Satellite Receiver 2025 – 2026
Kubernetes, Python, FastAPI, Vue.js, GNU Radio, ZeroMQ

- Owned the end-to-end Kubernetes application architecture, connecting the PlutoSDR handler and GNU Radio sinks to the backend, frontend, and dynamically allocated audio and spectrogram workers.
- Delivered a live QO-100 receiver with real-time audio and waterfall visualization in collaboration with teams from Padua, Télécom Saint-Étienne, and TU Darmstadt.

SQL over LLMs – Reproducibility Study Oct 2025 – Jan 2026
Python, DuckDB, SQLGlot, FastAPI, query optimization

- Reproduced and evaluated the SIGMOD 2025 Galois framework and documented implementation choices, experimental results, and differences from the original work.
- Implemented a query engine that uses SQLGlot to transform SQL into logical and physical execution plans containing LLM-backed data-retrieval operators.

Traveling Salesman Problem Solver Mar 2025 – Feb 2026
C, IBM CPLEX, combinatorial optimization

- Implemented and evaluated constructive heuristics, local search, and metaheuristics in C; integrated IBM CPLEX for Branch-and-Cut, Benders Decomposition, Hard Fixing, and Local Branching methods.

SEUPD@CLEF: Team BASETTE at LongEval Mar 2025 – Jul 2025
First author; Java, Lucene, Python, Optuna, information retrieval

- Led the implementation of a multithreaded Lucene retrieval system designed for strong performance on commodity CPU hardware and co-authored the accepted CLEF 2025 working-notes paper.
- Profiled indexing and retrieval, then built an Optuna workflow to tune BM25, preprocessing, reranking, and filtering parameters for retrieval effectiveness and stability across time-shifted corpora.

TECHNICAL SKILLS

Core	C++, C, Python, MLIR/LLVM, compiler design, code generation, FIMDRAM/PIM, computer architecture
Performance	perf, Valgrind, ASan/UBSan, combinatorial optimization, IBM CPLEX, Branch-and-Cut, heuristics
Research and data	Java, Lucene, DuckDB, SQLGlot, query optimization, information retrieval, statistical analysis
Tools and platforms	Git, Linux, Docker, Kubernetes, CMake, GNU Radio, ZeroMQ, Proxmox
AI tools	AI coding assistants, Model Context Protocol (MCP), plugins
Additional	PHP/Laravel, Vue.js, TypeScript, Inertia.js, FastAPI, SQL

LANGUAGES

Languages	Italian (native), English (C1)